Object Oriented Programming (CSEG1044)

Unit 06 – Lecture 06: Testing + Build/Artifacts + Final Demo

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February 27, 2026

Sources: Git SCM: *Pro Git* (2e) (version control workflow), accessed February 27, 2026.; GitHub Docs (repositories, issues, pull requests), accessed February 27, 2026.; Oracle Java Tutorials: Swing + JDBC (accessed February 27, 2026).; JUnit 5 User Guide (accessed February 27, 2026).; Apache Maven Project: Getting Started (accessed February 27, 2026).; Gradle User Manual: Getting Started (accessed February 27, 2026).; UML class diagrams: UML 2.x overview/spec (accessed February 27, 2026).

Quick Links

Overview Key Topics Demo Summary

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Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

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Learning Outcomes

- Write JUnit tests for service/business rules.

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Learning Outcomes

- Write JUnit tests for service/business rules.
- Run an integration checklist for DAO + DB.

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Learning Outcomes

- Write JUnit tests for service/business rules.
- Run an integration checklist for DAO + DB.
- Package a runnable artifact and prepare a final demo (Deliverable-5).

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Goal: demo-ready project

- Core flows are reliable (not “works only sometimes”).

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Goal: demo-ready project

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- Errors are handled gracefully (validation + messages).

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Goal: demo-ready project

- Core flows are reliable (not "works only sometimes").
- Errors are handled gracefully (validation + messages).
- Build produces a runnable artifact (Jar/zip).

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Goal: demo-ready project

- Core flows are reliable (not "works only sometimes").
- Errors are handled gracefully (validation + messages).
- Build produces a runnable artifact (Jar/zip).
- You have a short demo script and clear evidence of teamwork.

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Deliverable-5 (final submission)

- Runnable artifact + run instructions.

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Deliverable-5 (final submission)

- Runnable artifact + run instructions.
- `schema.sql` + DB setup notes.

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Deliverable-5 (final submission)

- Runnable artifact + run instructions.
- `schema.sql` + DB setup notes.
- Unit tests (JUnit) for service rules.

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Deliverable-5 (final submission)

- Runnable artifact + run instructions.
- `schema.sql` + DB setup notes.
- Unit tests (JUnit) for service rules.
- Short report + README + team contributions.

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Unit 06: Capstone Project

- Lecture focus: Testing + Build/Artifacts + Final Demo

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Today: Roadmap

- Testing strategy: what to unit test vs integrate

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Today: Roadmap

- Testing strategy: what to unit test vs integrate
- JUnit 5 essentials (assertions + test structure)

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Today: Roadmap

- Testing strategy: what to unit test vs integrate
- JUnit 5 essentials (assertions + test structure)
- Build + packaging + demo checklist (Deliverable-5)

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Testing strategy (capstone)

- **Unit tests (fast):** service validation + business rules.

Rule: if a bug can be caught by a unit test, write the unit test.

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Testing strategy (capstone)

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- **Integration tests:** DAO + DB schema + mapping.

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Testing strategy (capstone)

- **Unit tests (fast):** service validation + business rules.
- **Integration tests:** DAO + DB schema + mapping.
- **Manual demo:** Swing flows (minimal but polished).

Rule: if a bug can be caught by a unit test, write the unit test.

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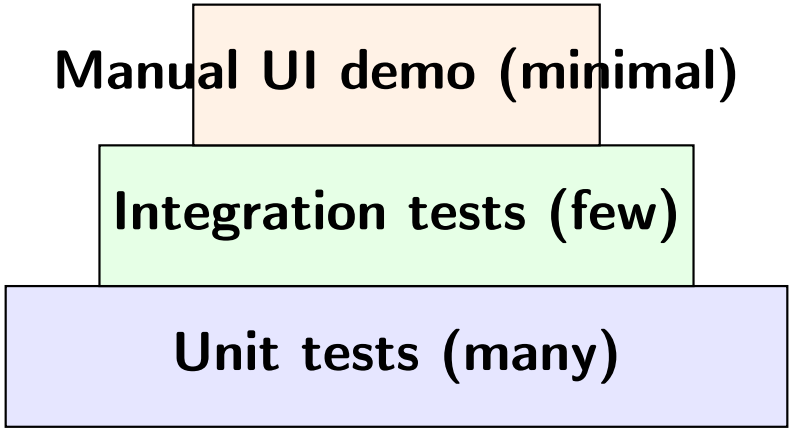
Test pyramid (in this course)



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JUnit 5 essentials

- Use `@Test` methods; follow Arrange–Act–Assert.

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JUnit 5 essentials

- Use `@Test` methods; follow Arrange–Act–Assert.
- Assertions: `assertEquals`, `assertTrue`, `assertThrows`.

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JUnit 5 essentials

- Use `@Test` methods; follow Arrange–Act–Assert.
- Assertions: `assertEquals`, `assertTrue`, `assertThrows`.
- Use `@BeforeEach` to set up a clean fixture.

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Example: assertThrows for validation

```
@Test
void rejectsEndBeforeStart() {
    assertThrows(ValidationException.class,
        () -> service.createBooking(1, "A", 10, 10));
}
```

- Good tests are small and check one behavior.

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- Name tests like a sentence: `rejectsEndBeforeStart`.

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Design for testability: interface + fake

- Tests run fast without DB.

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Design for testability: interface + fake

- Tests run fast without DB.
- Production uses JDBC implementation.

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Integration checklist (DAO + DB)

1. Run `schema.sql` on a clean DB.

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Integration checklist (DAO + DB)

1. Run `schema.sql` on a clean DB.
2. Insert one row using DAO.

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Integration checklist (DAO + DB)

1. Run `schema.sql` on a clean DB.
2. Insert one row using DAO.
3. List/find and verify mapping fields.

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Integration checklist (DAO + DB)

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2. Insert one row using DAO.
3. List/find and verify mapping fields.
4. Update and verify changes.
5. Delete and verify removal.

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Build + packaging (idea)

- Maven: `mvn test`, then `mvn package` (Jar in `target/`)

Tip: record your run steps in README (so TA can run it quickly).

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Build + packaging (idea)

- Maven: `mvn test`, then `mvn package` (Jar in `target/`)
- Gradle: `gradle test`, then `gradle build`

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Build + packaging (idea)

- Maven: `mvn test`, then `mvn package` (Jar in `target/`)
- Gradle: `gradle test`, then `gradle build`
- Artifact should be runnable with clear instructions.

Tip: record your run steps in README (so TA can run it quickly).

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Release + demo preparation

- Freeze scope: choose what you will demo.

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Release + demo preparation

- Freeze scope: choose what you will demo.
- Run tests + integration checklist.

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Release + demo preparation

- Freeze scope: choose what you will demo.
- Run tests + integration checklist.
- Prepare demo data (1–2 example resources, bookings).

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Release + demo preparation

- Freeze scope: choose what you will demo.
- Run tests + integration checklist.
- Prepare demo data (1–2 example resources, bookings).
- Create a release (tag) and attach the artifact.

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Demo script (5–7 mins)

1. Start DB; confirm schema exists.

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Demo script (5–7 mins)

1. Start DB; confirm schema exists.
2. Run the app; show one end-to-end flow (Add Resource → list).

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Demo script (5–7 mins)

1. Start DB; confirm schema exists.
2. Run the app; show one end-to-end flow (Add Resource → list).
3. Show booking flow (create + list).

Sources: Git SCM: *Pro Git* (2e) (version control workflow), accessed February 27, 2026.; GitHub Docs (repositories, issues, pull requests), accessed February 27, 2026.; Oracle Java Tutorials: Swing + JDBC (accessed February 27, 2026).; JUnit 5 User Guide (accessed February 27, 2026).; Apache Maven Project: Getting Started (accessed February 27, 2026).; Gradle User Manual: Getting Started (accessed February 27, 2026).; UML class diagrams: UML 2.x overview/spec (accessed February 27, 2026).

Demo script (5–7 mins)

1. Start DB; confirm schema exists.
2. Run the app; show one end-to-end flow (Add Resource → list).
3. Show booking flow (create + list).
4. Show validation (invalid booking shows a clean message).

Sources: Git SCM: *Pro Git* (2e) (version control workflow), accessed February 27, 2026.; GitHub Docs (repositories, issues, pull requests), accessed February 27, 2026.; Oracle Java Tutorials: Swing + JDBC (accessed February 27, 2026).; JUnit 5 User Guide (accessed February 27, 2026).; Apache Maven Project: Getting Started (accessed February 27, 2026).; Gradle User Manual: Getting Started (accessed February 27, 2026).; UML class diagrams: UML 2.x overview/spec (accessed February 27, 2026).

Demo script (5–7 mins)

1. Start DB; confirm schema exists.
2. Run the app; show one end-to-end flow (Add Resource → list).
3. Show booking flow (create + list).
4. Show validation (invalid booking shows a clean message).
5. (Optional) Run tests: `mvn test`.

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Mini Exercises (5 mins)

1. Write one unit test for a service rule in your project.

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Mini Exercises (5 mins)

1. Write one unit test for a service rule in your project.
2. What is one integration test you must do for DAO?

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Mini Exercises (5 mins)

1. Write one unit test for a service rule in your project.
2. What is one integration test you must do for DAO?
3. What 3 steps should be in your README "How to run"?

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Key Takeaways

- Test service rules with JUnit; keep tests fast and focused.

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Key Takeaways

- Test service rules with JUnit; keep tests fast and focused.
- Verify DAO with an integration checklist.

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Key Takeaways

- Test service rules with JUnit; keep tests fast and focused.
- Verify DAO with an integration checklist.
- Package a runnable artifact and rehearse a short demo script.

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Checkpoint Questions

1. What is the difference between unit and integration tests?

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Checkpoint Questions

1. What is the difference between unit and integration tests?
2. Why does "depend on an interface" help testing?

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Checkpoint Questions

1. What is the difference between unit and integration tests?
2. Why does "depend on an interface" help testing?
3. What must be included in a demo-ready release artifact?

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Course Wrap-Up

- Use OOP principles to keep code clean and reusable.

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Course Wrap-Up

- Use OOP principles to keep code clean and reusable.
- Use interfaces and layering to reduce coupling.

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Course Wrap-Up

- Use OOP principles to keep code clean and reusable.
- Use interfaces and layering to reduce coupling.
- Use testing + packaging to ship reliable software.

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